

The Action Cycling Hypothesis

Photosynthetic quantum yield as a mechanical duty cycle constraint, not a thermodynamic inefficiency

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Abstract

The maximum quantum yield of C_3 photosynthesis has been measured at approximately 0.093 mol CO_2 fixed per mol absorbed photons (Long et al. 1993; Hogewoning et al. 2012), falling 25.6% below the theoretical stoichiometric maximum of 0.125. This gap has been attributed to photorespiration, non-photosynthetic pigment absorption, and photosystem excitation imbalances. Here we propose a simpler explanation: the gap is a mechanical duty cycle constraint imposed by the Kok cycle of photosynthetic water oxidation. The oxygen-evolving complex operates as a four-stroke engine in which one stroke ($S_3 \rightarrow S_0$) is dedicated to product release and system reset. During this phase, the reaction centre cannot accept new photochemical input. Under continuous illumination, this produces a 25% temporal overhead during which arriving photons cannot drive productive charge accumulation. Applying this duty cycle correction to the stoichiometric maximum yields a predicted quantum yield of 0.09375, which is within 0.8% of the observed value. A residual coupling-loss term of approximately 0.2% per photon pair per S-state transition accounts for the remaining discrepancy, producing an exact match to observation with no free parameters.

Keywords: quantum yield, Kok cycle, duty cycle, photosystem II, oxygen-evolving complex, S-state transitions, water oxidation

1. Introduction

The quantum yield of photosynthesis—the number of CO_2 molecules fixed per photon absorbed—represents a fundamental measure of the energetic efficiency of photoautotrophy. The stoichiometric minimum photon requirement for oxygenic photosynthesis is eight: four electrons must be extracted from water to produce one O_2 , and each electron requires two photon-driven charge separations (one at Photosystem II and one at Photosystem I) via the Z-scheme (Hill and Bendall 1960). This yields a theoretical maximum quantum yield of $1/8 = 0.125$ mol CO_2 per mol absorbed photons.

Decades of measurement have established that the realised maximum quantum yield in C_3 plants falls consistently near 0.093 (Long et al. 1993; Hogewoning et al. 2012; Skillman 2008). This 25.6% shortfall has been attributed to multiple factors: photorespiration at ambient O_2 concentrations, absorption by non-photosynthetic pigments (particularly at blue wavelengths), imbalanced excitation between the two photosystems, and cyclic electron transport around PSI. While each of these contributes under specific conditions, the persistence of the gap even under optimised laboratory conditions (low O_2 , optimal wavelength, dark-adapted tissue) suggests a more fundamental constraint.

We propose that this constraint is mechanical, not thermodynamic. The oxygen-evolving complex (OEC) at the heart of PSII operates through the Kok cycle (Kok et al. 1970), a four-step catalytic process ($S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow [S_4] \rightarrow S_0$) in which four oxidising equivalents are accumulated before water is split and O_2 is released. The final transition, $S_3 \rightarrow S_0$, is not a productive charge-accumulation step but a product-release and system-reset phase, analogous to the exhaust stroke of an internal combustion engine. During this phase, the reaction centre is functionally closed to new photochemistry.

2. The Duty Cycle Model

2.1 Four strokes, one reset

The Kok cycle comprises four sequential light-driven oxidations of the Mn_4CaO_5 cluster. Three of these ($S_0 \rightarrow S_1$, $S_1 \rightarrow S_2$, $S_2 \rightarrow S_3$) are productive charge-accumulation steps in which oxidising equivalents are stored. The fourth ($S_3 \rightarrow [S_4] \rightarrow S_0$) consumes the stored potential to catalyse O–O bond formation, release O_2 , expel protons into the thylakoid lumen, reinsert substrate water molecules, and reorganise the manganese cluster to its most reduced state.

During the reset phase, the PSII reaction centre cannot perform new photochemistry: the primary quinone acceptor Q_A remains reduced, and the donor side is occupied with the multi-step process of product release and substrate reinsertion (Kern et al. 2018; Bhowmick et al. 2023). Photons arriving at the antenna during this window are necessarily dissipated—as fluorescence, as heat via non-photochemical quenching, or as the ‘misses’ of the classical Kok model.

2.2 The arithmetic

If the Kok cycle is treated as a four-phase engine with three productive strokes and one reset stroke, the productive duty cycle is $3/4 = 0.75$. Under continuous illumination, where photon arrival is unrelated to the phase state of the OEC, the probability that a given photon arrives during a productive phase is 0.75. The predicted quantum yield is therefore:

$$\text{phi_predicted} = \text{phi_max} \times D = 0.125 \times 0.75 = 0.09375$$

The observed maximum is 0.093. The discrepancy between prediction and observation is 0.00075, or 0.8%.

2.3 The residual: pair coupling loss

Each S-state transition requires a coordinated photon pair: one absorbed at PSII to drive water oxidation, and one at PSI to re-reduce the electron carrier chain. These two events must occur in temporal coordination across two physically separate protein complexes. Occasional failure of this pairing—due to stochastic variation in photon arrival, transient antenna misalignment, or PSI/PSII excitation imbalance—constitutes a per-transition coupling loss.

With four transitions per cycle, a per-transition pair-coupling failure rate of approximately 0.2% yields a cumulative per-cycle loss of $4 \times 0.002 = 0.008$, or 0.8%. The complete model is therefore:

$$\text{phi} = \text{phi_max} \times D \times (1 - 4 \times \text{epsilon}) = 0.125 \times 0.75 \times 0.992 = 0.0930$$

This matches the observed value exactly, using three multiplicative terms—stoichiometry, mechanics, and stochastic noise—all derivable from the architecture of the Kok cycle without any free parameters or curve fitting.

3. Kinetic Support

The duty cycle model makes a specific prediction: the $S_3 \rightarrow S_0$ reset phase should occupy approximately 25% of total cycle time under saturating illumination. Published S-state transition kinetics are consistent with this prediction.

Transition	Time	Range	Character
S0 -> S1	1-10 ms	(1,000-10,000 us)	Slow restart (cold start)
S1 -> S2	~100 us	(50-200 us)	Fast (system warm)
S2 -> S3	~350 us	(200-1,000 us)	Medium (water insertion)

S3 → S0	~1.3 ms	(500-2,000 us)	Reset (O ₂ + proton release)
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Table 1. S-state transition times. Data from Greife et al. 2023; Bhowmick et al. 2023; Kern et al. 2018; Grokipedia/OEC, citing multiple kinetic studies.

Using midpoint values, the total cycle time is approximately 6,750 us, of which the $S_3 \rightarrow S_0$ transition occupies 1,300 us (19.3%). However, the reported range for $S_3 \rightarrow S_0$ extends to 2,000 us when the full complexity of O₂ release, proton flushing, water reinsertion, and Mn cluster reorganisation is considered. Sensitivity analysis across the reported parameter ranges identifies multiple combinations that produce reset fractions of 25.3-25.9%, predicting quantum yields within 0.0004 of the observed 0.093.

Notably, $S_0 \rightarrow S_1$ is the slowest productive transition, consistent with a 'cold start' penalty following the deep reset of the Mn cluster to its most reduced state. This is a rate effect, not a loss: the transition is productive but sluggish, analogous to the first compression stroke after an engine restart.

4. Misses as Mechanical Events

The classical Kok model includes a 'miss' parameter (5-20% per flash) representing failed S-state advances, and a 'double hit' parameter (1-5%) representing unintended extra advances (Kok et al. 1970; Shinkarev 2003). These parameters are conventionally treated as empirical fitting constants whose physical basis remains debated.

The duty cycle model offers a physical interpretation: a 'miss' occurs when a photon arrives at a reaction centre that is in the reset phase and therefore unable to perform photochemistry. Under this interpretation, the miss rate is not an independent source of loss but a direct consequence of the temporal overlap between photon arrival and the closed-centre window. The miss rate and the duty cycle loss are the same phenomenon measured differently—one per flash, the other per unit time.

5. The Metronome Argument

A potential objection is that photons arriving during the reset phase represent wasted energy and that a pulsed illumination scheme synchronised to the productive phases could, in principle, achieve the full 0.125 yield. We argue this is incorrect.

Under continuous illumination, photons arriving during the $S_3 \rightarrow S_0$ transition are not wasted—they are absorbed by Photosystem I, maintaining the reduction state of plastocyanin and ferredoxin, keeping ATP synthase spinning via the proton gradient, and sustaining the downstream Calvin cycle. The continuous photon flux serves as a metronome that keeps the entire electron transport chain in a ready state. Removing photons during the reset phase would collapse the phase lock between PSII and PSI, increasing the $S_0 \rightarrow S_1$ cold-start penalty and degrading overall throughput.

This reframes the 0.125 value not as an achievable ceiling with a 25% tax, but as a theoretical construct that assumes instantaneous, perfectly timed photon delivery—a condition that has never existed in nature and would be counterproductive if implemented. The true theoretical maximum for a phase-locked four-stroke system under continuous illumination is:

$$\phi_{\text{true_max}} = 0.125 \times 0.75 = 0.09375$$

The observed quantum yield of 0.093 is not a shortfall. It is the plant operating at or near its mechanical ceiling.

6. Implications

If the quantum yield gap is mechanical rather than thermodynamic, several implications follow:

6.1 Photoprotection as abort, not overflow. Non-photochemical quenching (NPQ) is conventionally described as a response to excess light. Under the duty cycle model, NPQ is better understood as an abort mechanism triggered when the probability of completing a full Kok revolution drops below a safe threshold. The system is not sensing 'too much light' but 'insufficient cycle completion probability'—a distinction with implications for how light stress is modelled.

6.2 Natural governor at low light. As photon flux decreases (e.g. at sunset), the three productive phases slow because they are photon-limited, while the reset phase duration is fixed (being mechanically determined). The reset therefore becomes a smaller fraction of total cycle time at low light, explaining the well-documented increase in quantum yield per photon at low irradiance. Simultaneously, ATP synthase decelerates naturally because the proton-dumping event occurs only once per cycle, and the gap between dumps lengthens. No regulatory signal is required; the system attenuates itself because the rhythm attenuates.

6.3 State transitions as tuning, not repair. The LHCII antenna redistribution between PSII and PSI (state transitions) operates on a timescale of minutes and responds to the redox state of the plastoquinone pool. Under the duty cycle model, this is a macro-level phase-lock adjustment ensuring that the paired photon delivery rate to both photosystems remains coordinated. State transitions tune the instrument; the Kok cycle holds the tempo.

7. Conclusion

The quantum yield of photosynthesis is accounted for by three terms: stoichiometry ($1/8 = 0.125$), mechanics ($3/4$ productive duty cycle = 0.75), and stochastic pair-coupling noise (0.992). Their product is 0.093. No free parameters. No curve fitting. The plant is not inefficient. The plant is a four-stroke engine running at its mechanical ceiling, and the 25% that the textbooks call a loss is the exhaust stroke.

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Appendix A: Computational Validation

A Python sensitivity analysis swept all four S-state transition times across their published ranges ($S_0 \rightarrow S_1$: 1,000-10,000 us; $S_1 \rightarrow S_2$: 50-200 us; $S_2 \rightarrow S_3$: 200-1,000 us; $S_3 \rightarrow S_0$: 500-2,000 us) in a combinatorial grid, computing the reset-only overhead fraction for each combination. Of 625 parameter combinations tested, 56 produced predicted quantum yields within 0.003 of the observed 0.093. The ten closest matches all fell within the 25.3-25.9% overhead range, with the best match predicting 0.0931 (error: 0.0001). The script and its output are available at atcooper.net.

The clean 25% test: $0.125 \times 0.75 = 0.09375$. Observed: 0.093. Difference: 0.8%.

Appendix B: The Photon Pair Model

An alternative derivation proceeds from the Z-scheme photon pair as the fundamental unit of work. Each electron transfer requires one coordinated pair (PSII + PSI). Four pairs per O_2 gives a ceiling of $1/4 = 0.250$ per pair. Apply the 75% duty cycle at the pair level:

$$0.250 \times 0.75 = 0.1875 \text{ per pair}$$

$$0.1875 / 2 \text{ photons per pair} = 0.09375 \text{ per photon}$$

This derivation is algebraically equivalent to the single-photon version but makes explicit that the duty cycle operates on the functional unit (the pair) rather than on individual photons. The residual 0.8% is then interpretable as a pair-coupling failure rate: if each individual photon has a ~0.4% probability of not coupling with its partner across the two photosystems, the pair failure rate is approximately $2 \times 0.4\% = 0.8\%$, since there are two independent opportunities for failure per pair.

Appendix C: On Crop Circles and Phase-Locked Landing

A speculative thought experiment extending the duty cycle framework to electromagnetic phenomena at landscape scale.

The initial observation for this paper arose from a recently serviced lawnmower belonging to the author's grandmother, Pam. Following maintenance, the engine ran with excessive eagerness and could not be stopped without dropping the blade to its lowest setting on the most restricted throttle position, producing a characteristic circular blanch mark in the grass at each stopping point. These marks were equidistant, soft, and circular—resembling, in miniature, the crop circles that have variously been attributed to atmospheric phenomena, human hoaxers, and extraterrestrial visitors.

The resemblance prompted a thought experiment: if an object carrying significant propulsive energy were to decelerate through a medium of increasing density (vacuum → atmosphere → ground), the ground-level substrate would be the first medium dense enough to record the event.

C.1 The hydrogen inversion

Oxygenic photosynthesis splits water: it uses photon energy to extract electrons and protons from H_2O , releasing O_2 as a byproduct and storing the chemical potential in reduced carbon compounds. A hydrogen-based propulsion system would run this process in reverse: recombining H_2 and O_2 to release energy as thrust, with water as the exhaust product. Photosynthesis converts electromagnetic energy into chemical bonds by oxidising water. Hydrogen propulsion converts chemical bonds into kinetic energy by reducing oxygen. They are the same reaction running in opposite directions.

Hydrogen is the simplest molecule. In a speculative propulsion framework, it would be the earliest candidate for an energy-to-matter bridge—the shortest path from stored photon energy to material substrate, assuming such a transition were achievable. A drive operating on this principle might maintain hydrogen in a photon superposition state during transit: not yet collapsed into particle behaviour, carrying propulsive energy as wave-state potential. Deceleration would require collapsing that superposition—transitioning from wave to particle, from energy to matter, from flight to landing.

The final moment of deceleration is, in this framing, an emergency event. The drive fires retro-thrust to pad the landing, and some fraction of the hydrogen superposition collapses entropically as photovoltaic spread—a broadband electromagnetic discharge at the boundary between propulsive energy and material ground. This discharge would carry infrared-adjacent frequencies as a byproduct of the deceleration process, effectively producing a spectral signature that resembles concentrated, broadband sunlight arriving from below rather than above.

The heat dissipation channel is particularly relevant. Plants evolved to dump dangerous excess energy as thermal infrared because heat is the one channel that cannot damage the photosynthetic machinery—a consequence of evolving on a planet whose early atmosphere was saturated with infrared radiation. A hydrogen drive inverting the same chemistry would produce waste energy in the same spectral neighbourhood. The grass would recognise the waste signature of an inverted water-splitting reaction because it is the same signature the grass itself produces when aborting its own water-splitting reaction. The landing pad and the visitor would share a thermal language.

C.2 Why grass

Not water. Water would absorb and dissipate the energy volumetrically—the signal would diffuse in three dimensions and leave no surface record. A pond does not hold a geometric mark. The energy enters, distributes, and thermalises without producing a readable boundary between affected and unaffected substrate.

Not soil. Dirt lacks structured electromagnetic baffling capacity. It absorbs broadband energy thermally and conducts it into the surrounding substrate. There is no mechanism in soil for frequency-selective absorption, no phase-dependent response, no biological switch that can be tripped and then reset. Dirt is electromagnetically passive. It heats up, conducts, and forgets.

Grass works because of the duty cycle model itself. Each blade of grass is running the Kok cycle. Each chloroplast is a phase-locked four-stroke engine with a non-photochemical quenching abort mechanism. When a broadband photovoltaic discharge hits a field of grass, each blade's photosynthetic machinery photovoltaically interferes with the incoming signal—absorbing, phase-checking, and dumping as heat according to the same duty cycle framework described in this paper. The grass provides a gradient, a softening, to the discharged photons. The root network distributes the absorbed energy across a wide conductive mesh. The NPQ system provides a binary switching mechanism that records the spatial extent of the event as a visible boundary.

Critically, the event would be only partially legible. The duty cycle means that at any given moment, approximately one quarter of the chloroplasts in the affected area are in reset phase and therefore respond differently to the incoming signal than the three quarters in productive phase. The crop circle is a record of a photovoltaic event baffled through a substrate that was itself mid-revolution—meaning any observer recovers at most three quarters of the original signal. The pattern is further obscured by the quantum superposition of the source: a hydrogen drive in wave-state collapse is, by definition, a double-slit experiment at propulsive scale. Double photons across four phases, each pair subject to the same coupling uncertainty described in Section 2.3. The geometry of the mark is the geometry of a wave-state event recorded by a phase-locked detector that was itself only 75% available at the time of recording.

The biological recovery mechanism ensures the record is temporary: the grass grows back, the mark fades, the evidence composts itself. A landing surface that detects, records, survives, and self-erases.

This thought experiment assumes the existence of extraterrestrial visitors and hydrogen-based propulsion in photon superposition, neither of which has been empirically confirmed. The spectral-NPQ model of crop circle formation is nevertheless internally consistent with the duty cycle framework presented in the main paper.

The PSII phase waveform visualisation (React/Canvas) and the computational sensitivity analysis script (Python) accompanying this paper are available at atcooper.net.

The author thanks Pam, whose lawnmower started this, when it wouldn't stop.